

PYONGGANG, Korea, Democratic People's Republic of

WMO: 470750

Lat: 38.417N

Lon: 127.283E

Elev: 1155

StdP: 14.09

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.6	(c) 3.5	(d) -8.0	(e) 3.7	(f) 3.4	(g) -4.4	(h) 4.5	(i) 7.1	(j) 18.3	(k) 16.6	(l) 16.5	(m) 20.0	(n) 2.8	(o) 230	(p) 0.413

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.4	(c) 86.2	(d) 75.8	(e) 83.6	(f) 74.0	(g) 81.4	(h) 72.4	(i) 77.9	(j) 83.9	(k) 76.4	(l) 81.3	(m) 75.1	(n) 78.9	(o) 4.6	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.1	142.1	81.5	74.9	136.5	79.3	73.8	131.5	77.8	42.4	84.2	40.9	81.7	39.6	79.2	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	15.5	13.4	-5.8	90.5	5.2	2.8	-9.5	92.5	-12.5	94.2	-15.4	95.8	-19.1	97.9
			-5.8	80.1	5.1	1.7	-9.5	81.3	-12.5	82.3	-15.4	83.3	-19.1	84.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	19.9	25.5	36.6	48.4	59.9	68.2	73.3	74.1	64.4	53.1	38.9	24.0
	DBStd	19.88	8.14	7.57	6.68	6.89	5.08	3.81	4.04	4.40	4.80	6.08	8.32	8.48
	HDD50	3340	934	686	419	111	2	0	0	0	0	39	342	807
	HDD65	6560	1399	1106	881	499	168	15	1	2	66	368	783	1272
	CDD50	2968	0	0	3	62	309	545	724	746	434	136	9	0
	CDD65	715	0	0	0	0	11	109	260	284	50	1	0	0
	CDH74	4121	0	0	0	8	188	640	1318	1748	212	7	0	0
	CDH80	838	0	0	0	0	16	101	284	431	6	0	0	0
(14) Wind	WSAvg	4.5	4.0	4.5	5.6	6.0	5.5	4.6	4.5	3.4	3.4	3.9	4.6	4.7
(15) Precipitation	PrecAvg	55.7	0.7	1.0	1.2	2.6	4.1	5.5	17.6	12.9	5.4	2.1	1.9	1.0
	PrecMax	88.9	1.9	3.9	3.0	6.3	12.8	15.8	45.9	36.2	16.7	8.7	5.4	3.1
	PrecMin	26.9	0.0	0.0	0.2	0.4	0.6	2.0	6.1	1.7	0.3	0.2	0.4	0.2
	PrecStd	13.0	0.5	0.9	0.7	1.4	2.5	3.5	9.1	8.3	4.1	1.9	1.1	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.1	50.6	62.5	74.9	82.0	86.0	89.9	90.3	80.8	74.3	62.9	46.0
		MCWB	37.1	43.9	49.6	56.4	63.0	70.1	77.7	79.2	68.8	62.1	54.4	42.7
	2%	DB	37.2	44.3	56.9	70.0	78.5	82.8	85.5	87.2	78.6	70.5	58.0	42.2
		MCWB	33.3	36.4	47.4	53.6	61.4	68.7	76.1	77.4	68.4	59.7	51.2	38.7
	5%	DB	34.7	40.4	52.4	66.1	75.7	80.3	82.8	84.5	76.6	67.2	54.3	39.0
		MCWB	31.8	34.7	44.7	51.9	60.3	68.0	74.6	76.2	66.9	57.7	48.4	35.8
	10%	DB	32.5	37.0	48.3	62.2	72.2	77.7	80.4	82.0	74.4	64.5	51.4	36.1
		MCWB	29.8	32.5	41.3	50.3	59.1	66.6	73.8	74.7	65.8	56.1	46.9	33.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.7	46.5	52.8	60.1	67.0	73.3	79.3	80.4	73.8	64.9	56.3	43.4
		MCDB	40.4	51.0	58.1	71.2	75.6	81.1	86.2	88.6	77.5	71.1	60.7	45.3
	2%	WB	34.4	38.3	48.7	56.3	64.3	71.3	77.4	78.5	71.1	61.5	52.3	39.1
		MCDB	36.3	42.2	55.5	64.9	72.7	78.9	83.3	84.9	75.2	67.3	56.1	41.7
	5%	WB	32.2	35.2	45.3	54.0	62.5	69.8	76.0	77.2	68.9	59.4	49.8	36.2
		MCDB	34.3	39.4	51.6	62.9	71.6	76.8	80.7	82.7	74.0	65.4	53.5	38.5
	10%	WB	30.1	32.7	41.9	51.7	60.8	68.4	74.9	75.9	67.2	57.5	47.3	33.7
		MCDB	32.5	36.5	47.5	60.5	69.4	74.8	78.7	80.3	72.6	63.2	50.8	36.0
(35) Mean Daily Temperature Range	MDBR		18.3	17.9	17.3	21.2	20.5	17.0	11.3	13.4	18.3	20.7	16.5	15.9
	5% DB	MCDBR	17.6	20.8	24.0	27.1	27.2	22.1	16.0	16.9	21.0	24.0	20.2	16.1
		MCWBR	14.1	14.3	14.8	13.5	12.4	10.2	7.9	8.4	11.0	13.7	13.4	12.9
	5% WB	MCDBR	15.3	17.3	22.5	23.6	21.5	17.7	12.6	14.5	15.9	20.8	16.7	15.3
		MCWBR	13.1	12.9	14.8	13.6	11.6	9.5	6.8	7.8	9.5	13.1	12.5	12.9
(40) Clear-Sky Solar Irradiance	taub		0.391	0.454	0.533	0.592	0.597	0.662	0.634	0.543	0.474	0.475	0.446	0.391
	taud		1.966	1.850	1.743	1.671	1.700	1.626	1.747	1.971	2.080	1.954	1.959	2.011
	Ebn at Noon		233	234	228	225	227	213	218	235	244	225	213	223
	Edh at Noon		42	54	67	76	75	81	71	56	47	48	43	38
(44) All-Sky Solar Radiation	RadAvg		771	1038	1318	1550	1738	1623	1245	1316	1285	1106	752	687
	RadStd		58	69	129	129	172	152	183	173	144	79	84	60

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+185	+97

Nomenclature: See separate page